

2016.0 RANGE ROVER (LG), 501-25

BODY REPAIRS – GENERAL INFORMATION

DESCRIPTION AND OPERATION

Bonding

This section provides information on the bonded joints used in repair throughout the all new L405 aluminium body. The following topics are covered

1 Equipment

2 Materials

3 Bonding Pre Treatment

4 Bonding Application

It is a pre-requisite that any person undertaking any repairs which involve panel bonding has attended Land Rover Approved Training and has achieved the correct skill level to undertake these processes. Bonding is classed as a Category 'A' repair.



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Equipment

The approved bonding equipment consists of:

- 1 Pyrosil gas applicator kit.
- 2 Approved two-pack adhesive applicator.

Bonding Pre-treatment

Pre-treating the panels to be bonded, as described in this section, promotes the improved adhesion of the bonding adhesive.

The pre-treatment is carried out using the approved "Pyrosil kit". The kit contains all of the equipment required to perform the pre-treatment process.



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CAUTION:

Bonding without pre-treatment will reduce the performance of the joint.

The pre-treatment application is a two stage process:

- 1** Chemical application via a flame.
- 2** Coupling Agent.

Performing the Pre-treatment Application

CAUTION:

Apply caution when using a naked flame. Remove all risk of combustion. Do not overheat the alloy. Move the flame over the work piece at a constant velocity.

The application is colorless and has no visual indication.

- 1** Remove any original adhesive, or E-Coat from new panels.
- 2** Clean surface to be treated using approved pre-cleaner/degreaser.
- 3** Using the Pyrosil torch, pass the flame, (use the blue tip which is the oxidization flame), across the surface of the joint.
- 4** Immediately brush coupling agent (primer) onto panel surface and allow to dry.

CAUTION:

Apply the bonding adhesive immediately after the pre-treatment process.

Bonding Application

The approved bonding adhesive is supplied in a two pack cartridge.

A two-pack cartridge gun with a disposable mixing nozzle is used for the application of the adhesive.



NOTE:

The curing process of the adhesive begins once mixed and has a working time of approximately 30 minutes.

Apply a 5mm "zig zag" bead directly to either the replacement panel or vehicle structure.

Make sure a continuous bead of adhesive surrounds fixing holes.

On completion the mixing nozzle must be discarded. Make sure an air tight seal is provided for the cartridges.

